Looking more closely at Earth's atmospheric fingerprints, human biomarkers will also include sulfuric, carbonic, and nitric acids, and other components of smog from the burning of fossil fuels. If the curious aliens happen to be socially, culturally, and technologically more advanced than we are, then they will surely interpret these biomarkers as convincing evidence for the absence of intelligent life on Earth.

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The first exoplanet was discovered in 1995, and, as of this writing, the tally is rising through three thousand, most found in a small pocket of the Milky Way around the solar system. So there's plenty more where they came from. After all, our galaxy contains more than a hundred billion stars, and the known universe harbors some hundred billion galaxies.

Our search for life in the universe drives the search for exoplanets, some of which resemble Earth—not in detail, of course, but in overall properties. Latest estimates, extrapolating from the current catalogs, suggests as many as forty billion Earth-like planets in the Milky Way alone. Those are the planets our descendants might want to visit someday, by choice, if not by necessity.

† Jocelyn Bell, *Annals of the New York Academy of Sciences* 302 (1977): 685.